

FairShare-GPU: Dual-Platform Experiment Plan

This document outlines a focused comparison of GPU sharing techniques across two distinct hardware platforms: NVIDIA A100 (for core mechanism comparison) and NVIDIA T4 (for extensive scheduling sweeps).

Common Workload: A heterogeneous batch of 10 jobs (5 LLM Inference + 5 General ML Jobs).

1. NVIDIA A100 Experiments: MIG vs. MPS Comparison

Objective: Test the core trade-off between Hard Isolation and Shared Utilization.

Phase 1: A100 Baselines

A1 — JRT Baseline & Setup

- Sharing Mode: None / MIG
- Goal: Establish isolated JRT for all 10 jobs and verify MIG configuration.

Phase 2: A100 Isolation Comparison (MIG vs. MPS)

A2 — MPS Unconstrained Contention

- Sharing Mode: MPS (mps_level=100)
- Config: 5 jobs (Group A) + 5 jobs (Group B)
- Focus: Maximum interference, highest P99 latency.

A3 — MPS Constrained Isolation

- Sharing Mode: MPS (mps_level=50)
- Same setup as A2.
- Focus: Benefit of explicit software control, improved fairness.

A4 — MIG Hard Isolation

- Sharing Mode: MIG (e.g., two 2g.10gb slices)

- Focus: Minimum interference, best predictable P99 latency.

2. NVIDIA T4 Experiments: MPS Scheduling Sweeps

Objective: Test impact of workload ordering and resource control on time-sharing performance.

Phase 1: T4 Scheduling and Ordering

T1 — Random Order (Baseline)

- MPS (mps_level=100)
- Focus: Baseline JCT and Makespan.

T2 — Worst-Case Order

- Largest jobs first → smallest last
- MPS (mps_level=100)
- Focus: Max HOL blocking, lowest fairness.

T3 — Fair/Optimal Order

- Smallest jobs first → largest last
- MPS (mps_level=100)
- Focus: Best achievable JCT and Makespan.

Phase 2: T4 Resource Allocation Sweep

T4 — Resource Allocation Sweep

- Worst-case ordering
- Sweep mps_level = 25, 33, 50, 100
- Focus: Fairness vs. Throughput tradeoff curve.

Core Project Deliverables

Scheduling Impact

- Source: T4 (T1, T2, T3)

- Metric: Jain's Fairness Index, Queue Time.

MIG vs. MPS for Inference

- Source: A100 (A2, A3, A4)

- Metric: P99 Latency.

Utilization vs. Predictability

- Source: A100 (A2, A4)

- Metrics: GPU Utilization vs. Interference Factor.